Web Technology

Unit VI: Server Side Scripting using PHP

Class Lecture
+
Lab Session
8 Hours

6.1. PHP Syntax

6.2.Variables, Data Types , Strings, Constants

6.3.Operators, Control structure, Functions, Array

6.4.Creating Class and Objects

6.5.PHP Forms, Accessing Form Elements, Form Validation, Events

6.6.Cookies and Sessions

6.7.Working with PHP and MySQL, Connecting to Database, Creating, Selecting, Deleting, Updating Records in a table, Inserting Multiple Data

6.8. Introduction to CodeIgniter, Laravel, Wordpress (Basic concepts and Installation)

What is PHP

- PHP is an open-source, interpreted, and object-oriented scripting language that can be executed at the server-side. PHP is well suited for web development.
- Therefore, it is used to develop web applications (an application that executes on the server and generates the dynamic page.).
- PHP was created by **Rasmus Lerdorf in 1994** but appeared in the market in 1995. **PHP 7.4.0** is the latest version of PHP, which was released on **28 November**.

Some important points need to be noticed about PHP are as followed

- PHP stands for Hypertext Preprocessor.
- PHP is an interpreted language, i.e., there is no need for compilation.
- PHP is faster than other scripting languages, for example, ASP and JSP.
- PHP is a server-side scripting language, which is used to manage the dynamic content of the website.
- PHP can be embedded into HTML.
- PHP is an object-oriented language.
- PHP is an open-source scripting language.
- PHP is simple and easy to learn language.

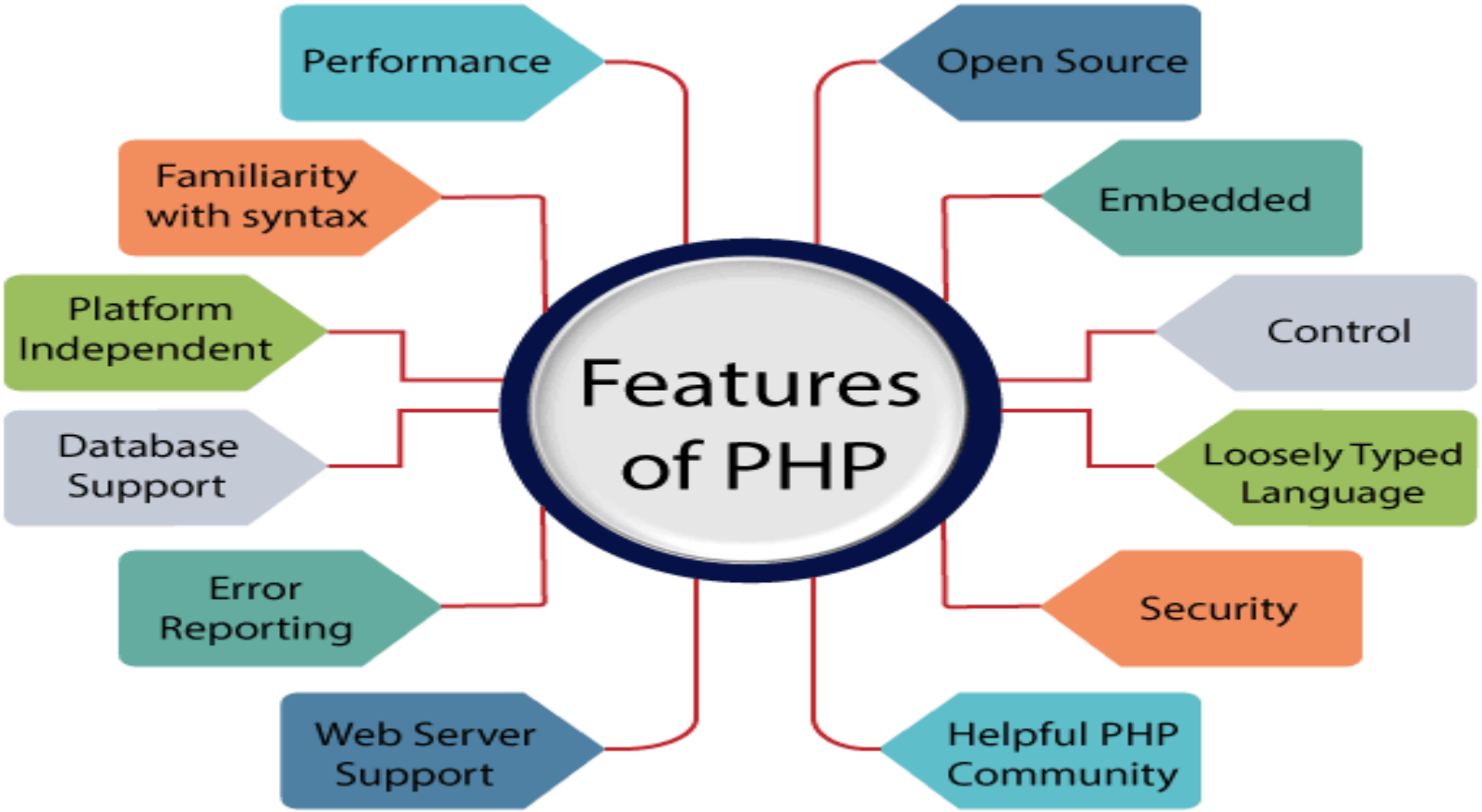
Why use PHP

PHP is a server-side scripting language, which is used to design the dynamic web applications with MySQL database.

- It handles dynamic content, database as well as session tracking for the website.
- You can create sessions in PHP.
- It can access cookies variable and also set cookies.
- It helps to encrypt the data and apply validation.
- PHP supports several protocols such as HTTP, POP3, SNMP, LDAP, IMAP, and many more.
- Using PHP language, you can control the user to access some pages of your website.
- As PHP is easy to install and set up, this is the main reason why PHP is the best language to learn.
- PHP can handle the forms, such as - collect the data from users using forms, save it into the database, and return useful information to the user. For example - Registration form.

PHP Features

PHP is very popular language because of its simplicity and open source. There are some important features of PHP given below:



Install PHP

To install PHP, we will suggest you to install AMP (Apache, MySQL, PHP) software stack. It is available for all operating systems. There are many AMP options available in the market that are given below:

- WAMP for Windows
- LAMP for Linux
- MAMP for Mac
- SAMP for Solaris
- FAMP for FreeBSD
- XAMPP (Cross, Apache, MySQL, PHP, Perl) for Cross Platform: It includes some other components too such as FileZilla, OpenSSL, Webalizer, Mercury Mail, etc.

➢ If you are on Windows and don't want Perl and other features of XAMPP, you should go for WAMP. In a similar way, you may use LAMP for Linux and MAMP for Macintosh.

PHP Case Sensitivity

- In PHP, keyword (e.g., echo, if, else, while), functions, user-defined functions, classes are not case-sensitive. However, all variable names are case-sensitive.

- In the below example, you can see that all three echo statements are equal and valid:

Example-1

```
<!DOCTYPE>
<html>
   <body>
      <?php
         echo "Hello world using echo </br>";
         ECHO "Hello world using ECHO </br>";
         EcHo "Hello world using EcHo </br>";
      ?>
   </body>
</html>
```

Class Demo for PHP Running Code

Open xampp folder
open Htdoc
make Folder and open with it by visual studio
Make file name index.php and type code php
Browser –localhost/name of folder
Index:-self running program

- Look at the below example that the variable names are case sensitive. You can see the example below that only the second statement will display the value of the $color variable. Because it treats $color, $ColoR, and $COLOR as three different variables:

```
<html>
   <body>
      <?php
         $color = "black";
         echo "My car is ". $ColoR ."</br>";
         echo "My dog is ". $color ."</br>";
         echo "My Phone is ". $COLOR ."</br>";
      ?>
   </body>
</html>
```

Only $color variable has printed its value, and other variables $ColoR and $COLOR are declared as undefined variables. An error has occurred in line 5 and line 7.

PHP Echo

- PHP echo is a language construct, not a function. Therefore, you don't need to use parenthesis with it. But if you want to use more than one parameter, it is required to use parenthesis.

- The syntax of PHP echo is given below:

```
void echo ( string $arg1 [, string $... ] )
```

- PHP echo statement can be used to print the string, multi-line strings, escaping characters, variable, array, etc. Some important points that you must know about the echo statement are:

- echo is a statement, which is used to display the output.
- echo can be used with or without parentheses: echo(), and echo.
- echo does not return any value.
- We can pass multiple strings separated by a comma (,) in echo.
- echo is faster than the print statement.

PHP echo: printing string

```
<?php
echo "Hello by PHP echo";
?>
```

PHP echo: printing multi line string

```
<?php
echo "Hello by PHP echo
this is multi line
text printed by
PHP echo statement
";
?>
```

PHP echo: printing escaping characters

File: echo3.php

```
<?php
echo "Hello escape \"sequence\" characters";
?>
```

PHP echo: printing variable value

File: echo4.php

```
<?php
$msg="Hello JavaTpoint PHP";
echo "Message is: $msg";
?>
```

Practical Demo

PHP Print

- Like PHP echo, PHP print is a language construct, so you don't need to use parenthesis with the argument list. Print statement can be used with or without parentheses: print and print(). Unlike echo, it always returns 1.

  The syntax of PHP print is given below:

  int print(string $arg)

- PHP print statement can be used to print the string, multi-line strings, escaping characters, variable, array, etc. Some important points that you must know about the echo statement are:
  - print is a statement, used as an alternative to echo at many times to display the output.
  - print can be used with or without parentheses.
  - print always returns an integer value, which is 1.
  - Using print, we cannot pass multiple arguments.
  - print is slower than the echo statement.

PHP print: printing string

File: print1.php

```
<?php
print "Hello by PHP print ";
print ("Hello by PHP print()");
?>
```

PHP print: printing multi line string

File: print2.php

```
1.<?php
2.print "Hello by PHP print
3.this is multi line
4.text printed by
5.PHP print statement
6.";
7.?>
```

PHP print: printing escaping characters

File: print3.php

```
<?php
print "Hello escape \"sequence\" characters by PHP print";
1.?>
```

PHP print: printing variable value

File: print4.php

```
1.<?php
2.$msg="Hello print() in PHP";
3.print "Message is: $msg";
4.?>
```

6.2.Variables, Data Types , Strings, Constants

❑ PHP Variables

- In PHP, a variable is declared using a $ sign followed by the variable name. Here, some important points to know about variables:
- As PHP is a loosely typed language, so we do not need to declare the data types of the variables. It automatically analyzes the values and makes conversions to its correct datatype.
- After declaring a variable, it can be reused throughout the code.
- Assignment Operator (=) is used to assign the value to a variable.

Syntax of declaring a variable in PHP is given below:

```
$variablename=value;
```

❑ Rules for declaring PHP variable:

1. A variable must start with a dollar ($) sign, followed by the variable name.
2. It can only contain alpha-numeric character and underscore (A-z, 0-9, _).
3. A variable name must start with a letter or underscore (_) character.
4. A PHP variable name cannot contain spaces.
5. One thing to be kept in mind that the variable name cannot start with a number or special symbols.
6. PHP variables are case-sensitive, so $name and $NAME both are treated as different variable.

❑ PHP Variable: Declaring string, integer, and float

Let's see the example to store string, integer, and float values in PHP variables.

File: variable1.php

```
<?php
$str="hello string";
$x=200;
$y=44.6;
echo "string is: $str <br/>";
echo "integer is: $x <br/>";
echo "float is: $y <br/>";
?>
```

❑ PHP Variable: Sum of two variables

File: variable2.php

```
<?php
$x=5;
$y=6;
$z=$x+$y;
echo $z;
?>
```

PHP Variable: case sensitive

In PHP, variable names are case sensitive. So variable name "color" is different from Color, COLOR, COLor etc.

File: variable3.php

```
<?php
$color="red";
echo "My car is " . $color . "<br>";
echo "My house is " . $COLOR . "<br>";
echo "My boat is " . $coLOR . "<br>";
?>
```

PHP Variable: Rules

PHP variables must start with letter or underscore only.

PHP variable can't be start with numbers and special symbols.

File: variablevalid.php

```
<?php
$a="hello";//letter (valid)
$_b="hello";//underscore (valid)

echo "$a <br/> $_b";
?>
```

❑ PHP Variable Scope

The scope of a variable is defined as its range in the program under which it can be accessed. In other words, "The scope of a variable is the portion of the program within which it is defined and can be accessed." PHP has three types of variable scopes:

1.Local variable
2.Global variable
3.Static variable

Local variable

The variables that are declared within a function are called local variables for that function. These local variables have their scope only in that particular function in which they are declared. This means that these variables cannot be accessed outside the function, as they have local scope.

File: local_variable1.php

```
1.<?php
2.    function local_var()
3.    {
4.        $num = 45;  //local variable
5.        echo "Local variable declared inside the function is: ". $num;
6.    }
7.    local_var();
8.?>
```

File: local_variable2.php

```
<?php
    function mytest()
    {
        $lang = "PHP";
        echo "Web development language: " .$lang;
    }
    mytest();
    //using $lang (local variable) outside the function wil
l generate an error
    echo $lang;
?>
```

Global variable

The global variables are the variables that are declared outside the function. These variables can be accessed anywhere in the program. To access the global variable within a function, use the GLOBAL keyword before the variable. However, these variables can be directly accessed or used outside the function without any keyword. Therefore there is no need to use any keyword to access a global variable outside the function.

Note: Without using the global keyword, if you try to access a global variable inside the function, it will generate an error that the variable is undefined.

File: global_variable1.php

```
1.<?php
2.    $name = "Sanaya Sharma";        //Global Variable
3.    function global_var()
4.    {
5.        global $name;
6.        echo "Variable inside the function: ". $name;
7.        echo "</br>";
8.    }
9.    global_var();
10.    echo "Variable outside the function: ". $name;
11.?>
```

```
<?php
    $name = "Sanaya Sharma";        //global variable
    function global_var()
    {
        echo "Variable inside the function: ". $name;

        echo "</br>";
    }
    global_var();
?>
```

Static variable

It is a feature of PHP to delete the variable, once it completes its execution and memory is freed. Sometimes we need to store a variable even after completion of function execution. Therefore, another important feature of variable scoping is static variable. We use the static keyword before the variable to define a variable, and this variable is called as **static variable**.

Static variables exist only in a local function, but it does not free its memory after the program execution leaves the scope. Understand it with the help of an example:

Example:

File: static_variable.php

```
<?php
   function static_var()
   {
      static $num1 = 3;       //static variable
      $num2 = 6;         //Non-static variable
      //increment in non-static variable
      $num1++;
      //increment in static variable
      $num2++;
      echo "Static: " .$num1 ."</br>";
      echo "Non-static: " .$num2 ."</br>";
   }

//first function call
   static_var();

   //second function call
   static_var();
?>
```

You have to notice that $num1 regularly increments after each function call, whereas $num2 does not. This is why because $num1 is not a static variable, so it freed its memory after the execution of each function call.

PHP Constants

- PHP constants are name or identifier that can't be changed during the execution of the script except for magic constants, which are not really constants. PHP constants can be defined by 2 ways:
  1. Using define() function
  2. Using const keyword
- Constants are similar to the variable except once they defined, they can never be undefined or changed. They remain constant across the entire program. PHP constants follow the same PHP variable rules. **For example**, it can be started with a letter or underscore only.
- Conventionally, PHP constants should be defined in uppercase letters.

Note: Unlike variables, constants are automatically global throughout the script.

File: constant1.php

```
<?php
define("MESSAGE","Hello JavaTpoint PHP");
echo MESSAGE;
?>
```

File: constant4.php

```
<?php
const MESSAGE="Hello const by JavaTpoint PHP";
echo MESSAGE;
?>
```

Constant	Variables
Once the constant is defined, it can never be redefined.	A variable can be undefined as well as redefined easily.
A constant can only be defined using define() function. It cannot be defined by any simple assignment.	A variable can be defined by simple assignment (=) operator.
There is no need to use the dollar ($) sign before constant during the assignment.	To declare a variable, always use the dollar ($) sign before the variable.
Constants do not follow any variable scoping rules, and they can be defined and accessed anywhere.	Variables can be declared anywhere in the program, but they follow variable scoping rules.
Constants are the variables whose values can't be changed throughout the program.	The value of the variable can be changed.
By default, constants are global.	Variables can be local, global, or static.

PHP Data Types

PHP data types are used to hold different types of data or values. PHP supports 8 primitive data types that can be categorized further in 3 types:

1.Scalar Types (predefined)
2.Compound Types (user-defined)
3.Special Types

PHP Data Types: Scalar Types

It holds only single value. There are 4 scalar data types in PHP.

1.boolean
2.integer
3.float
4.string

PHP Data Types: Compound Types

It can hold multiple values. There are 2 compound data types in PHP.

1.array
2.object

PHP Data Types: Special Types

There are 2 special data types in PHP.

1. resource
2. NULL

PHP Boolean

Booleans are the simplest data type works like switch. It holds only two values: **TRUE (1)** or **FALSE (0)**. It is often used with conditional statements. If the condition is correct, it returns TRUE otherwise FALSE.

Example:

```
<?php
   if (TRUE)
      echo "This condition is TRUE.";
   if (FALSE)
      echo "This condition is FALSE.";
?>
```

PHP Integer

Integer means numeric data with a negative or positive sign. It holds only whole numbers, i.e., numbers without fractional part or decimal points.

Rules for integer:

- An integer can be either positive or negative.
- An integer must not contain decimal point.
- Integer can be decimal (base 10), octal (base 8), or hexadecimal (base 16).
- The range of an integer must be lie between 2,147,483,648 and 2,147,483,647 i.e., -2^31 to 2^31.

Example:

```
<?php
   $dec1 = 34;
   $oct1 = 0243;
   $hexa1 = 0x45;
   echo "Decimal number: " .$dec1. "</br>";
   echo "Octal number: " .$oct1. "</br>";
   echo "HexaDecimal number: " .$hexa1. "</br>";
?>
```

Example: floating point

```
<?php
   $n1 = 19.34;
   $n2 = 54.472;
   $sum = $n1 + $n2;
   echo "Addition of floating numbers: " .$sum;
?>
```

Example Array

```
<?php
    $bikes = array ("Royal Enfield", "Yamaha", "KTM");
   var_dump($bikes);   //the var_dump() function returns the datatype and values
   echo "</br>";
   echo "Array Element1: $bikes[0] </br>";
   echo "Array Element2: $bikes[1] </br>";
   echo "Array Element3: $bikes[2] </br>";
?>
```

PHP Operators

PHP Operators can be categorized in following forms:

- Arithmetic Operators
- Assignment Operators
- Bitwise Operators
- Comparison Operators
- Incrementing/Decrementing Operators
- Logical Operators
- String Operators
- Array Operators
- Type Operators
- Execution Operators
- Error Control Operators

```
<?php
   //class declaration
   class Developer
   {}
   class Programmer
   {}
   //creating an object of type Developer
   $charu = new Developer();

   //testing the type of object
   if( $charu instanceof Developer)
   {
      echo "Charu is a developer.";
   }
   else
   {
      echo "Charu is a programmer.";
   }
   echo "</br>";
   var_dump($charu instanceof Developer);          //It will return true.
   var_dump($charu instanceof Programmer);       //It will return false.
?>
```

PHP If Else

PHP if else statement is used to test condition. There are various ways to use if statement in PHP.

- if
- if-else
- if-else-if
- nested if

```php
<?php
$num=12;
if($num%2==0){
echo "$num is even number";
}else{
echo "$num is odd number";
}
?>
```

```php
<?php
        $age = 23;
   $nationality = "Indian";
   //applying conditions on nationality and age
   if ($nationality == "Indian")
   {
      if ($age >= 18) {
         echo "Eligible to give vote";
      }
      else {
         echo "Not eligible to give vote";
      }
   }
?>
```

```php
<?php
   $marks=69;
   if ($marks<33){
      echo "fail";
   }
   else if ($marks>=34 && $marks<50) {
      echo "D grade";
   }
   else if ($marks>=50 && $marks<65) {
     echo "C grade";
   }
   else if ($marks>=65 && $marks<80) {
      echo "B grade";
   }
   else if ($marks>=80 && $marks<90) {
      echo "A grade";
   }
   else if ($marks>=90 && $marks<100) {
      echo "A+ grade";
   }
  else {
      echo "Invalid input";
   }
?>
```

PHP Switch

PHP switch statement is used to execute one statement from multiple conditions. It works like PHP if-else-if statement.

Important points to be noticed about switch case:

1.The **default** is an optional statement. Even it is not important, that default must always be the last statement.
2.There can be only one **default** in a switch statement. More than one default may lead to a **Fatal** error.
3.Each case can have a **break** statement, which is used to terminate the sequence of statement.
4.The **break** statement is optional to use in switch. If break is not used, all the statements will execute after finding matched case value.
5.PHP allows you to use number, character, string, as well as functions in switch expression.
6.Nesting of switch statements is allowed, but it makes the program more complex and less readable.
7.You can use semicolon (;) instead of colon (:). It will not generate any error.

```
1.<?php
2.$num=20;
3.switch($num){
4.case 10:
5.echo("number is equals to 10");
6.break;
7.case 20:
8.echo("number is equal to 20");
9.break;
10.case 30:
11.echo("number is equal to 30");
12.break;
13.default:
14.echo("number is not equal to 10, 20 or 30");
15.}
16.?>
```

```php
<?php
    $ch = 'U';
    switch ($ch)
    {
        case 'a':
            echo "Given character is vowel";
            break;
        case 'e':
            echo "Given character is vowel";
            break;
        case 'i':
            echo "Given character is vowel";
            break;
        case 'o':
            echo "Given character is vowel";
            break;
        case 'u':
            echo "Given character is vowel";
            break;
        case 'A':
            echo "Given character is vowel";
            break;
        case 'E':
            echo "Given character is vowel";
            break;
        case 'I':
            echo "Given character is vowel";
            break;
        case 'O':
            echo "Given character is vowel";
            break;
        case 'U':
            echo "Given character is vowel";
            break;
        default:
            echo "Given character is consonant";
            break;
    }
?>
```

6.7.Working with PHP and MySQL, Connecting to Database, Creating, Selecting, Deleting, Updating Records in a table, Inserting Multiple Data

❑ PHP MySQL Connect

Since PHP 5.5, mysql_connect() extension is deprecated. Now it is recommended to use one of the 2 alternatives.

1. mysqli_connect()
2. PDO::__construct()

	PDO	MySQLi
Database support	12 different drivers	MySQL only
API	OOP	OOP + procedural
Connection	Easy	Easy
Named parameters	Yes	No
Object mapping	Yes	Yes
Prepared statements (client side)	Yes	No
Performance	Fast	Fast
Stored procedures	Yes	Yes

Finally, MySqli is native for php and works very well with MySql. So, you can choose Mysqli only if you are working on Mysql database and not familiar with PDO statements. Its always recommended to prefer PDO, as it supports twelve different database drivers including Mysql database. Performance point of view Mysqli is better and security point of view, both are safe as long as the developer follows recommended statements.

❑ PHP mysqli_connect()

PHP mysqli_connect() function is used to connect with MySQL database. It returns resource if connection is established or null.

Syntax

resource mysqli_connect (server, username, password)

❑ PHP mysqli_close()

PHP mysqli_close() function is used to disconnect with MySQL database. It returns true if connection is closed or false.

Syntax

bool mysqli_close(resource $resource_link)

PHP MySQL Connect Example

Example

```
<?php
$host = 'localhost:3306';
$user = '';
$pass = '';
$conn = mysqli_connect($host, $user, $pass);
if(! $conn )
{
  die('Could not connect: ' . mysqli_error());
}
echo 'Connected successfully';
mysqli_close($conn);
?>
```

```php
<?php
// connect to data base we have define following
$severname ="localhost";// since we run in our computer
$username ="root";    // since no user name
$password ="" ;        // no password local server

// To connect server we need to do
$conn = mysqli_connect($servername,$username,$password);
// for message we use if and die function
if(!conn){
   die("sorry we failed to connect:" .mysqli_connect_error());
}
else {
   echo"connection was successfully";

}
?>
```

❑ Open a Connection to MySQL

Before we can access data in the MySQL database, we need to be able to connect to the server:

Example (MySQLi Object-Oriented)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";

// Create connection
$conn = new mysqli($servername, $username, $password);

// Check connection
if ($conn->connect_error) {
  die("Connection failed: " . $conn->connect_error);
}
echo "Connected successfully";
?>
```

Note on the object-oriented example above:

$connect_error was broken until PHP 5.2.9 and 5.3.0. If you need to ensure compatibility with PHP versions prior to 5.2.9 and 5.3.0, use the following code instead:

```
// Check connection
if (mysqli_connect_error()) {
  die("Database connection failed: " .
mysqli_connect_error());
}
```

❑ Example (MySQLi Procedural)

```php
<?php
$servername = "localhost";
$username = "username";
$password = "password";

// Create connection
$conn = mysqli_connect($servername,
$username, $password);

// Check connection
if (!$conn) {
  die("Connection failed: " .
mysqli_connect_error());
}
echo "Connected successfully";
?>
```

❑ Example (PDO)

```php
<?php
$servername = "localhost";
$username = "username";
$password = "password";

try {
  $conn
= new PDO("mysql:host=$servername;dbname
=myDB", $username, $password);
  // set the PDO error mode to exception
  $conn->setAttribute(PDO::ATTR_ERRMODE,
PDO::ERRMODE_EXCEPTION);
  echo "Connected successfully";
} catch(PDOException $e) {
  echo "Connection failed: " . $e-
>getMessage();
}
?>
```

Note: In the PDO example above we have also **specified a database (myDB)**. PDO require a valid database to connect to. If no database is specified, an exception is thrown.
Tip: A great benefit of PDO is that it has an exception class to handle any problems that may occur in our database queries. If an exception is thrown within the try{ } block, the script stops executing and flows directly to the first catch(){ } block.

Create a MySQL Database Using MySQLi and PDO

Create a MySQL Database Using MySQLi and PDO

Example (MySQLi Object-oriented) database name myDB

```
<?php
$servername = "localhost";
$username = “root";
$password = "";

// Create connection
$conn = new mysqli($servername, $username, $password);
// Check connection
if ($conn->connect_error) {
  die("Connection failed: " . $conn->connect_error);
}

// Create database
$sql = "CREATE DATABASE myDB";
if ($conn->query($sql) === TRUE) {
  echo "Database created successfully";
} else {
  echo "Error creating database: " . $conn->error;
}

$conn->close();
?>
```

Note: The following PDO example create a database named "myDBPDO":

Example (PDO)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";

try {
  $conn = new PDO("mysql:host=$servername", $username, $password);
  // set the PDO error mode to exception
  $conn->setAttribute(PDO::ATTR_ERRMODE,
PDO::ERRMODE_EXCEPTION);
  $sql = "CREATE DATABASE myDBPDO";
  // use exec() because no results are returned
  $conn->exec($sql);
  echo "Database created successfully<br>";
} catch(PDOException $e) {
  echo $sql . "<br>" . $e->getMessage();
}

$conn = null;
?>
```

PHP MySQL Create Table

Create a MySQL Table Using MySQLi and PDO
The CREATE TABLE statement is used to create a table in MySQL.

We will create a table named "MyGuests", with five columns: "id", "firstname", "lastname", "email" and "reg_date":

```
CREATE TABLE MyGuests (
id INT(6) UNSIGNED AUTO_INCREMENT PRIMARY KEY,
firstname VARCHAR(30) NOT NULL,
lastname VARCHAR(30) NOT NULL,
email VARCHAR(50),
reg_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
)
```

Notes on the table above:

The data type specifies what type of data the column can hold. For a complete reference of all the available data types, go to our Data Types reference.

After the data type, you can specify other optional attributes for each column:

NOT NULL - Each row must contain a value for that column, null values are not allowed
DEFAULT value - Set a default value that is added when no other value is passed
UNSIGNED - Used for number types, limits the stored data to positive numbers and zero
AUTO INCREMENT - MySQL automatically increases the value of the field by 1 each time a new record is added
PRIMARY KEY - Used to uniquely identify the rows in a table. The column with PRIMARY KEY setting is often an ID number, and is often used with AUTO_INCREMENT
Each table should have a primary key column (in this case: the "id" column). Its value must be unique for each record in the table.

Example (MySQLi Object-oriented)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";

// Create connection
$conn = new mysqli($servername, $username, $password, $dbname);
// Check connection
if ($conn->connect_error) {
  die("Connection failed: " . $conn->connect_error);
}

// sql to create table
$sql = "CREATE TABLE MyGuests (
id INT(6) UNSIGNED AUTO_INCREMENT PRIMARY KEY,
firstname VARCHAR(30) NOT NULL,
lastname VARCHAR(30) NOT NULL,
email VARCHAR(50),
reg_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON
UPDATE CURRENT_TIMESTAMP
)";

if ($conn->query($sql) === TRUE) {
  echo "Table MyGuests created successfully";
} else {
  echo "Error creating table: " . $conn->error;
}

$conn->close();
```

Example (MySQLi Procedural)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";

// Create connection
$conn = mysqli_connect($servername, $username, $password, $dbname);
// Check connection
if (!$conn) {
  die("Connection failed: " . mysqli_connect_error());
}

// sql to create table
$sql = "CREATE TABLE MyGuests (
id INT(6) UNSIGNED AUTO_INCREMENT PRIMARY KEY,
firstname VARCHAR(30) NOT NULL,
lastname VARCHAR(30) NOT NULL,
email VARCHAR(50),
reg_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP
)";

if (mysqli_query($conn, $sql)) {
  echo "Table MyGuests created successfully";
} else {
  echo "Error creating table: " . mysqli_error($conn);
}

mysqli_close($conn);
?>
```

Example (PDO)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDBPDO";

try {
  $conn = new PDO("mysql:host=$servername;dbname=$dbname", $username, $password);
  // set the PDO error mode to exception
  $conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);

  // sql to create table
  $sql = "CREATE TABLE MyGuests (
  id INT(6) UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  firstname VARCHAR(30) NOT NULL,
  lastname VARCHAR(30) NOT NULL,
  email VARCHAR(50),
  reg_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP
  )";

  // use exec() because no results are returned
  $conn->exec($sql);
  echo "Table MyGuests created successfully";
} catch(PDOException $e) {
  echo $sql . "<br>" . $e->getMessage();
}

$conn = null;
?>
```

❑ PHP MySQL Insert Data

Insert Data Into MySQL Using MySQLi and PDO
After a database and a table have been created, we can start adding data in them.

❑ Here are some syntax rules to follow:

- The SQL query must be quoted in PHP
- String values inside the SQL query must be quoted
- Numeric values must not be quoted
- The word NULL must not be quoted

The INSERT INTO statement is used to add new records to a MySQL table:

```
INSERT INTO table_name (column1, column2, column3,...)
VALUES (value1, value2, value3,...)
```

Note: If a column is AUTO_INCREMENT (like the "id" column) or TIMESTAMP with default update of current_timesamp (like the "reg_date" column), it is no need to be specified in the SQL query; MySQL will automatically add the value.

Example (MySQLi Object-oriented)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";

// Create connection
$conn = new mysqli($servername, $username, $password, $dbname);
// Check connection
if ($conn->connect_error) {
  die("Connection failed: " . $conn->connect_error);
}

$sql = "INSERT INTO MyGuests (firstname, lastname, email)
VALUES ('John', 'Doe', 'john@example.com')";

if ($conn->query($sql) === TRUE) {
  echo "New record created successfully";
} else {
  echo "Error: " . $sql . "<br>" . $conn->error;
}

$conn->close();
?>
```

Example (MySQLi Procedural)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";

// Create connection
$conn = mysqli_connect($servername, $username, $password,
$dbname);
// Check connection
if (!$conn) {
  die("Connection failed: " . mysqli_connect_error());
}

$sql = "INSERT INTO MyGuests (firstname, lastname, email)
VALUES ('John', 'Doe', 'john@example.com')";

if (mysqli_query($conn, $sql)) {
  echo "New record created successfully";
} else {
  echo "Error: " . $sql . "<br>" . mysqli_error($conn);
}

mysqli_close($conn);
?>
```

Example (PDO)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDBPDO";

try {
  $conn = new PDO("mysql:host=$servername;dbname=$dbname",
$username, $password);
  // set the PDO error mode to exception
  $conn->setAttribute(PDO::ATTR_ERRMODE,
PDO::ERRMODE_EXCEPTION);
  $sql = "INSERT INTO MyGuests (firstname, lastname, email)
  VALUES ('John', 'Doe', 'john@example.com')";
  // use exec() because no results are returned
  $conn->exec($sql);
  echo "New record created successfully";
} catch(PDOException $e) {
  echo $sql . "<br>" . $e->getMessage();
}

$conn = null;
?>
```

❑ Insert Multiple Records Into MySQL Using MySQLi and PDO

Multiple SQL statements must be executed with the mysqli_multi_query() function.

The following examples add three new records to the "MyGuests" table:

Example (MySQLi Object-oriented)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";

// Create connection
$conn = new mysqli($servername, $username, $password, $dbname);
// Check connection
if ($conn->connect_error) {
  die("Connection failed: " . $conn->connect_error);
}
```

```
$sql = "INSERT INTO MyGuests (firstname, lastname, email)
VALUES ('John', 'Doe', 'john@example.com');";
$sql .= "INSERT INTO MyGuests (firstname, lastname, email)
VALUES ('Mary', 'Moe', 'mary@example.com');";
$sql .= "INSERT INTO MyGuests (firstname, lastname, email)
VALUES ('Julie', 'Dooley', 'julie@example.com')";

if ($conn->multi_query($sql) === TRUE) {
  echo "New records created successfully";
} else {
  echo "Error: " . $sql . "<br>" . $conn->error;
}

$conn->close();
?>
```

Example (MySQLi Procedural)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";

// Create connection
$conn = mysqli_connect($servername, $username, $password, $dbname);
// Check connection
if (!$conn) {
  die("Connection failed: " . mysqli_connect_error());
}

$sql = "INSERT INTO MyGuests (firstname, lastname, email)
VALUES ('John', 'Doe', 'john@example.com');";
$sql .= "INSERT INTO MyGuests (firstname, lastname, email)
VALUES ('Mary', 'Moe', 'mary@example.com');";
$sql .= "INSERT INTO MyGuests (firstname, lastname, email)
VALUES ('Julie', 'Dooley', 'julie@example.com')";

if (mysqli_multi_query($conn, $sql)) {
  echo "New records created successfully";
} else {
  echo "Error: " . $sql . "<br>" . mysqli_error($conn);
}

mysqli_close($conn);
?>
```

Example (PDO)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDBPDO";

try {
  $conn = new PDO("mysql:host=$servername;dbname=$dbname", $username, $password);
  // set the PDO error mode to exception
  $conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);

  // begin the transaction
  $conn->beginTransaction();
  // our SQL statements
  $conn->exec("INSERT INTO MyGuests (firstname, lastname, email)
  VALUES ('John', 'Doe', 'john@example.com')");
  $conn->exec("INSERT INTO MyGuests (firstname, lastname, email)
  VALUES ('Mary', 'Moe', 'mary@example.com')");
  $conn->exec("INSERT INTO MyGuests (firstname, lastname, email)
  VALUES ('Julie', 'Dooley', 'julie@example.com')");

  // commit the transaction
  $conn->commit();
  echo "New records created successfully";
} catch(PDOException $e) {
  // roll back the transaction if something failed
  $conn->rollback();
  echo "Error: " . $e->getMessage();
}

$conn = null;
?>
```

❑ PHP MySQL Select Data

Select Data From a MySQL Database
The SELECT statement is used to select data from one or more tables:

SELECT column_name(s) FROM table_name
or we can use the * character to select ALL columns from a table:

SELECT * FROM table_name

Code lines to explain from the example above:

First, we set up an SQL query that selects the id, firstname and lastname columns from the MyGuests table. The next line of code runs the query and puts the resulting data into a variable called $result.

Then, the function num_rows() checks if there are more than zero rows returned.

If there are more than zero rows returned, the function fetch_assoc() puts all the results into an associative array that we can loop through. The while() loop loops through the result set and outputs the data from the id, firstname and lastname columns.

Select Data With MySQLi

The following example selects the id, firstname and lastname columns from the MyGuests table and displays it on the page:

Example (MySQLi Object-oriented)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";

// Create connection
$conn = new mysqli($servername, $username, $password, $dbname);
// Check connection
if ($conn->connect_error) {
  die("Connection failed: " . $conn->connect_error);
}

$sql = "SELECT id, firstname, lastname FROM MyGuests";
$result = $conn->query($sql);

if ($result->num_rows > 0) {
  // output data of each row
  while($row = $result->fetch_assoc()) {
    echo "id: " . $row["id"]. " - Name: " . $row["firstname"]. " " . $row["lastname"]. "<br>";
  }
} else {
  echo "0 results";
}
$conn->close();
?>
```

PHP MySQL Delete Data

The DELETE statement is used to delete records from a table:

```
DELETE FROM table_name
WHERE some_column = some_value
```

id	firstname	lastname	email	reg_date
1	John	Doe	john@example.com	2014-10-22 14:26:15
2	Mary	Moe	mary@example.com	2014-10-23 10:22:30
3	Julie	Dooley	julie@example.com	2014-10-26 10:48:23

Example (MySQLi Object-oriented)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";

// Create connection
$conn = new mysqli($servername, $username, $password,
$dbname);
// Check connection
if ($conn->connect_error) {
  die("Connection failed: " . $conn->connect_error);
}

// sql to delete a record
$sql = "DELETE FROM MyGuests WHERE id=3";

if ($conn->query($sql) === TRUE) {
  echo "Record deleted successfully";
} else {
  echo "Error deleting record: " . $conn->error;
}

$conn->close();
?>
```

The UPDATE statement is used to update existing records in a table:

```
UPDATE table_name
SET column1=value, column2=value2,...
WHERE some_column=some_value
```

```php
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";

// Create connection
$conn = new mysqli($servername, $username, $password, $dbname);
// Check connection
if ($conn->connect_error) {
  die("Connection failed: " . $conn->connect_error);
}

$sql = "UPDATE MyGuests SET lastname='Doe' WHERE id=2";

if ($conn->query($sql) === TRUE) {
  echo "Record updated successfully";
} else {
  echo "Error updating record: " . $conn->error;
}

$conn->close();
?>
```